Sparse Retrieval and BM25

1 Keyword retrieval

BM25 is a ranking function that scores a passage by the frequency of the query terms in it and their rarity across the corpus. TF-IDF is a weighting scheme that multiplies how often a term occurs in a passage by how rare the term is in the corpus. An inverted index is a structure that maps each term to the passages that contain it.

2 Behaviour

BM25 matches exact terms, so it handles acronyms and rare names well. Sparse retrieval fails when the query and the passage use different words for the same idea. BM25 needs no training and no vector index. Dense retrieval is a technique that maps text to vectors and ranks passages by their similarity to the query.